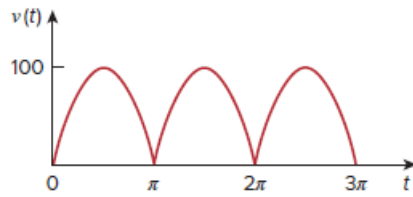


Problem

Find the rms value of the full-wave rectified sine wave in Fig. 11.17. Calculate the average power dissipated in a $6\text{-}\Omega$ resistor.

Figure 11.17:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.17 in the textbook for a full-wave rectified sine wave.

The time period of the waveform is $T = \pi$.

Write the expression for the voltage waveform.

$$v(t) = 100 \sin t, \quad 0 < t < \pi$$

Calculate the root mean square (rms) value of the waveform.

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v(t)^2 dt}$$

Substitute the expression for the voltage in the equation of V_{rms} .

$$\begin{aligned} V_{\text{rms}} &= \sqrt{\frac{1}{\pi} \int_0^{\pi} (100 \sin t)^2 dt} \\ &= \sqrt{\frac{1}{\pi} \int_0^{\pi} 10^4 \sin^2 t dt} \end{aligned}$$

Step 2 of 3

Consider the relation, $\sin^2 t = \frac{1}{2}(1 - \cos 2t)$ and use this in the calculation. Therefore,

$$\begin{aligned} V_{\text{rms}} &= \sqrt{\left(\frac{1}{\pi}\right)\left(\frac{10^4}{2}\right) \int_0^{\pi} (1 - \cos 2t) dt} \\ &= \sqrt{\frac{10^4}{2\pi} \left(t - \frac{\sin 2t}{2}\right) \Big|_0^{\pi}} \\ &= \sqrt{\frac{10^4}{2\pi} \left[(\pi - 0) - \left(\frac{\sin 2\pi}{2} - 0\right)\right]} \\ &= \sqrt{\frac{10^4 \pi}{2\pi}} \end{aligned}$$

Simplify the calculation further.

$$\begin{aligned} V_{\text{rms}} &= \sqrt{\frac{10^4}{2}} \\ &= \sqrt{5000} \\ &= 70.71 \text{ V} \end{aligned}$$

Thus, the rms value of the voltage waveform V_{rms} is 70.71 V.

Step 3 of 3

Calculate the average power dissipated in a 6Ω resistor.

$$P = \frac{V_{\text{rms}}^2}{R}$$

Substitute 70.71 V for V_{rms} and 6Ω for R .

$$\begin{aligned} P &= \frac{(70.71 \text{ V})^2}{6 \Omega} \\ &= \frac{5000}{6} \\ &= 833.34 \text{ W} \end{aligned}$$

Thus, the average power dissipated in a 6Ω resistor P is 833.34 W.